FIELD MANUAL

Master Chess

Turn your own games into your training plan.

DIAGNOSE ◆ EXPLAIN ◆ PRESCRIBE

An adaptive educational chess tutor

Player & Mechanics Manual • Version 1.0

How to read this manual

This manual is written the way a good board-game rulebook is written: the rules first, then a fully worked *game of play* that walks one real game from upload to a finished training plan, so you can see every mechanism act on real moves. Wherever you see a tinted box, it is doing one of three jobs.

EXAMPLE OF PLAY — HOW THESE BOXES WORK

Boxes like this one follow a single running example — the analysis of a real historical game — so that every abstract rule is immediately grounded in something concrete. Read them in order and you will have watched the whole system operate end to end.

■ Rule box — the exact mechanic

Boxes like this state a precise rule: a threshold, a formula, a piece of scheduling logic. These are the authoritative numbers, taken directly from the system's source of record.

★ NOTE

Boxes like this flag something easy to get wrong, a limitation, or a cross-reference worth keeping in mind.

THE ONE IDEA TO KEEP

Master Chess is not an engine with a chat window bolted on. It runs a **closed learning loop** — diagnose what you can and cannot do, explain it against engine truth, prescribe exactly what to study and drill — and **every score it shows you has a receipt**: the exact moves that produced it. No black box.

Contents

1 What Master Chess Is

Master Chess is an adaptive educational chess tutor. You upload your own games; it analyses every move with the Stockfish engine, and from that engine-grounded analysis it builds a persistent, *explainable* model of your ability across a taxonomy of **27 skills**. It then diagnoses your playing **level** and your current **plateau**, writes you a personalised **training plan** with matched reading, and turns the very mistakes you made into spaced-repetition **drills**.

The product's thesis is that expert chess knowledge should be encoded into a structured substrate, and instruction rendered dynamically for the individual learner from engine truth — never from an engine's chat-layer guesswork.

1.1 The two learning loops (and why the AI is neither)

The system has exactly two places where "learning" happens, and a language model is not one of them:

1. **The Player Model** — a structured representation of you across 27 skills, updated by Bayesian Knowledge Tracing from engine-verified evidence in every game and drill.
2. **The Chess Knowledge Layer** — engine ground truth, an opening / master-game library, and a book corpus mapped to the skill taxonomy.

An **AI reasoning layer** sits on top to write prose explanations, but it is always subordinate to engine truth: a runtime *claim guard* rejects any generated sentence that contradicts a verified move or evaluation. If the model cannot say something true, the app falls back to a deterministic, engine-only readout rather than let an unverified sentence reach you.

■ What grounds every claim

The engine (Stockfish) and the classified move record are the source of truth. The player model is derived from them by fixed, inspectable rules. The language model may only *phrase* facts it is handed; it may never introduce a move, number, or verdict of its own. Any sentence that fails verification is discarded.

1.2 How it runs

Layer	Technology
Client	React single-page app served by Vite (development port `5173`)
Server	Express API on `tsx` (default port `8030`)
Storage	SQLite via Drizzle ORM (`better-sqlite3`, WAL mode)
Engine	Stockfish over UCI, pooled; default search depth `16`, multi-PV `4`
Reasoning	Provider-abstracted LLM (local Ollama by default; hosted models by tier)

To run it locally: `npm run dev` starts client and server together; the server first applies database migrations, seeds the master-game library, resumes any interrupted analyses, and purges expired sessions. A health probe lives at `GET /api/health`.

> ★ **SCOPE NOTE**
>
> This manual documents the **Master Chess** standalone app (the `ai_chess` project). A sibling project in the same workspace is a separate classroom product with teacher/student dashboards; those screens are *not* part of Master Chess and are not described here.

2 The Vocabulary: Skills, Levels, Plateaus

Everything the app says about you is expressed in three linked vocabularies. Learn these and the rest of the manual reads easily.

2.1 The 27 skills, in 4 categories

Each skill carries a mastery score from 0–100, a trend arrow, and — once there is evidence — a list of the exact moves that produced the score. Three skills marked *partial* are only partially diagnosable from game data alone.

#	Skill	Diagnosable
OPENING		
1	Opening principles	yes
2	Opening repertoire	yes
3	Opening preparation & novelties	partial
MIDDLEGAME		
4	Strategic planning	yes
5	Positional play	yes
6	Patience & long-term thinking	yes
7	Prophylaxis	yes
8	Tactical pattern recognition	yes
9	Tactical consistency (blunder prevention)	yes
10	Calculation precision	yes
11	Attack & initiative	yes
12	Piece activity & coordination	yes
13	Defence & counterplay	yes
14	Converting advantages	yes
ENDGAME		

#	Skill	Diagnosable
15	Endgame principles	yes
16	Game transition (middlegame to endgame)	yes
17	Endgame precision & conversion	yes
18	Pawn endings	yes
19	Rook endings	yes
20	Knight endings	yes
21	Bishop & mixed-piece endings	yes
	PSYCHOLOGY & MENTAL	
22	Thought process & candidate moves	partial
23	Pattern recognition speed	yes
24	Intuition & practical judgment	partial
25	Time management	yes
26	Psychological resilience	yes
27	Psychological warfare	partial

> ★ **KNOWN COSMETIC DISCREPANCY**
>
> The taxonomy of record contains **27** skills, and the API always returns all 27. Some on-screen copy still reads “23 skills” — stale text from an earlier revision. The correct number is 27.

2.2 The 7 player levels

Your level is derived only from the rating carried in your games’ PGN headers (Chess.com and Lichess exports include it). Upload unrated games and the app honestly shows no level rather than guessing one.

ID	Name	Rating	Dominant failure mode
L1	Newcomer	0--799	Tactical invisibility --- misses what was just threatened
L2	Beginner	800--1199	Pattern blindness --- misses tactics beyond one move
L3	Casual club player	1200--1499	Contextual blindness --- misses in game what they solve in puzzles
L4	Improving club player	1500--1799	Strategic vacuum --- sound moves, no unifying plan
L5	Strong club player	1800--1999	Reactive play --- doesn't prevent the opponent's ideas
L6	Expert / CM	2000--2299	Conversion failure --- wins strategically, can't finish
L7	Master / Titled	2300+	Highly individual --- needs personalised diagnosis

2.3 The 5 plateaus

A plateau is a named stall pattern, auto-diagnosed from the *shape* of your skill vector — never self-reported. Zones deliberately overlap; ratings above 2200 have no formulaic plateau by design (that band is individual-coaching territory).

Plateau	Rating zone	Target skills
The blunder wall	0--1399	tactical pattern recognition, tactical consistency
The strategy desert	1400--1600	strategic planning, pawn endings, endgame principles
The conversion ceiling	1500--2000	converting advantages, rook endings, endgame precision
The prophylaxis gap	1700--1900	prophylaxis, defence & counterplay
The precision boundary	2000--2200	calculation precision, endgame precision, time mgmt, resilience

■ When a plateau is diagnosed

A plateau is only named when **(a)** your rating falls in its zone, **(b)** at least one of its target skills has real evidence, and **(c)** those target skills together carry at least **5** evidence samples. Among all qualifying plateaus, the one with the *lowest average mastery* across its evidenced targets wins. A brand-new account is given no diagnosis at all.

3 The Player's Journey

This section is the "how to play" walkthrough: the screens, in the order you meet them.

3.1 Register and sign in

Registration asks for a display name, an email, and a password of at least 8 characters. New accounts are created as role *player* on the *free* tier. Sign-in is rate-limited (20 attempts per 15 minutes per address) and returns a session token stored in your browser; it slides forward for 14 days of activity.

3.2 The Dashboard ("Games")

Your landing screen. Three stat cards summarise everything analysed so far — **Games analysed**, **Average accuracy**, and **Games with a blunder** — above a list of recent games. Each row shows the opening (ECO), both players with your side highlighted, the colour-coded result, time control, length, a blunder tag if any, and your accuracy. Accuracy is coloured green at 85+, gold at 75+, red below.

3.3 Upload games

Choose the source (Chess.com export, Lichess export, or manual PGN), tell the app **your username as it appears in the PGN** so it can detect which colour you played, then paste PGN text or drop a `.pgn` file. On submit, each game streams its analysis progress live. Unparseable games are listed with a reason and never sink the batch.

■ Upload limits & quotas

At most **50 games** per upload. Re-uploads of a game you already have are detected by content hash and cost nothing. Analysis is metered by tier: anonymous **1/day**, free **5/day**, Pro and Academy **unlimited**. Duplicate and over-quota games are never charged.

3.4 Game Review

The move-by-move board. On the left, an evaluation bar, the board oriented to your colour, and navigation (buttons or keyboard arrows / Home / End). Every move you step to shows a **classification badge**, the centipawns lost, how long you thought, the engine's preferred move, and the top three engine lines. On the right: live analysis progress, a **Game accuracy** card (White vs Black, with the engine depth and a tally of each classification), the full move list, and an **Engine summary** coaching note — either an LLM narrative badged "engine-grounded" or a deterministic fallback badged "deterministic".

3.5 Player Model

Headed *"Every score has a receipt. No black box."* On the left: your current level, your diagnosed plateau, an eight-axis skill radar, and a mastery-over-time sparkline. On the right: the full skill list with mastery bars and trend arrows, and an **Evidence** panel. Click any skill with evidence and it loads up to 20 receipts — the real moves behind the number, each marked as supporting the skill (✔) or counting against it (✘).

3.6 Training Plan

Headed *"Prescribed from your diagnosis, not a generic curriculum."* It opens with **The hypothesis** — a sentence tied to your diagnosed plateau — then up to three **focus blocks**. Each names a weak skill, its mastery and evidence count, a rationale, and **matched reading** drawn from a book corpus mapped to the taxonomy. A skill needs at least **3** evidence points before it is allowed into a plan.

3.7 Drills

Headed *"Every position here is one you actually reached and misplayed."* Each drill is one of your own mistake positions presented as a four-way multiple choice — the engine's best move plus three legal decoys. Answer correctly and the card is scheduled further out; answer wrongly and it returns tomorrow. A stat block tracks cards **due**, your day **streak**, and **retention**. The scheduler is SM-2 spaced repetition (§4).

3.8 Library

Two tabs. **Games** is a searchable catalogue of master games — the golden-start library ships with roughly **6,000** games from Morphy, Capablanca, Fischer, Tal, and Kasparov, and grows via `npm run library:import`. You can **search** by player, opening, or event, **filter** by ECO code, result, and source, **sort**, and page through the results. Loading a game runs full engine analysis on it for review (one quota unit); because a historical game has no honest "your side," a loaded classic is *not* folded into your player model — studying Morphy teaches you about Morphy, not about you. **Opening explorer** is an interactive tree: at every position it shows what the master library played there against what *you* have played there in your own analysed games — a direct read on where your repertoire leaks.

3.9 Account & data

GDPR controls. **Export** downloads every row keyed to you — games, moves, analyses, skill scores, evidence, coaching links — as a single JSON bundle (password fields stripped). **Delete** requires typing `DELETE`, then transactionally erases your data, tombstones the account, and logs you out. The deletion itself is recorded in an audit log for compliance.

4 The Mechanics

This is the rules reference: exactly how a game becomes a diagnosis. The worked example in §5 then runs all of it on one real game.

4.1 The pipeline, end to end

1. **Ingest.** The PGN is split (comment-safe), loaded move by move, and stored with before/after positions, clock times, and detected player colour.
2. **Evaluate.** Every position is scored by Stockfish (results are cached, since openings recur), producing an evaluation and ranked candidate lines for each.
3. **Classify.** Each move is labelled **Best** **Good** **Inaccuracy** **Mistake** **Blunder** by how much it lost against the engine's best, scaled by game phase.
4. **Infer evidence.** Fixed rules read the classified moves and emit "for" / "against" evidence for specific skills.
5. **Update the model.** Evidence is folded into each skill's Bayesian Knowledge Tracing state; a snapshot records level and plateau.
6. **Harvest & prescribe.** Mistakes become drills; the training plan is rebuilt; finally the narrator writes a claim-guarded summary.

> ★ **FAILURE ISOLATION**
>
> The engine analysis is committed and shown to you *before* the enrichment step (evidence, model, drills, narration) runs. Enrichment is wrapped so that if any part of it fails, your analysis still stands as "done" — and it is idempotent, so a restart mid-way never double-counts evidence.

4.2 Move classification

A move's raw loss is how many centipawns worse it is than the engine's best reply. Thresholds are **phase-scaled** — the endgame is judged more strictly than the opening, where many moves are near-equal:

Phase	good $>$	inaccuracy $\geq$	mistake $\geq$	blunder $\geq$
Opening	5	60	150	350
Middlegame	5	50	120	300
Endgame	5	40	90	200

Values are centipawns of *effective* loss. At or below ``good", a move is **Best**.

Phase detection is queen-aware

Phase is not just move number. Very low material (a lone minor per side or less, ≤ 500cp) forces **endgame** even early — an early queen trade into a king-and-pawn position is an endgame. Otherwise the first 20 plies are **opening**; after that, low non-pawn material *with no queens on the board* (≤ 1400cp) is **endgame**; everything else is **middlegame**. Queens keep a position out of the endgame even when other material is thin.

Win-probability dampening

Raw centipawns mislead in decided positions: going from +900 to +700 is not a blunder, because you were completely winning before and after. The classifier converts evaluations to win probability on the Lichess curve and, *only when both endpoints sit deep in a decided zone*, blends the raw loss toward the much smaller win-probability loss. A genuine thrown win — say +2000 collapsing to +200 — is *not* dampened, because the second endpoint is no longer decided. Missing a forced mate is always at least a mistake regardless of centipawns.

■ Accuracy score

Per move, the app computes win-probability points lost, maps each to a 0–100 accuracy on the Lichess curve ($103.1668\,e^{-0.04354x} - 3.1669$), then blends the arithmetic and harmonic means of the whole game. The harmonic term is what makes a single game-losing blunder cost more than a plain average would.

4.3 Turning moves into evidence

Rules fire only on *your* moves and emit evidence with a direction and a weight in $[0, 1]$. Error severity is graded blunder $= 1$, mistake $= 0.6$, inaccuracy $= 0.3$. A representative selection:

Rule	Fires when	Skill	Dir / weight
missed-forced-mate	you had a forced mate and gave it up	calculation precision	against / 1.0
opening-error	opening-phase mistake or blunder	opening principles	against / sev.
middlegame-blunder	a blunder in the middlegame	tactical consistency	against / 1.0
…-blunder-pattern	(same move, also)	tactical pattern recog.	against / 0.5
hanging-piece-left	error that leaves a piece capturable & undefended	piece activity	against / 0.4
endgame-error	endgame mistake or blunder	endgame precision	against / sev.
snap-blunder	error played in ≤ 3s	time management	against / 0.4
conversion-failure	reached $\geq +400$cp but didn't win	converting advantages	against / 0.8
sustained-initiative	grew an even game to $\geq +500$ cleanly	attack & initiative	for / 0.55
held-worse-position	were ≤ -300 yet didn't lose	defence & counterplay	for / 0.5
consistent-repertoire	same opening, played cleanly, ≥ 3 games	opening repertoire	for / 0.4
sound-novelty	sound engine-approved deviation from book	opening prep.	for / 0.3
prophylaxis probe	a null-move probe reveals a real threat	prophylaxis	for/against

★ **THE PROPHYLAXIS PROBE**

To judge prophylaxis, the app performs a "null-move" probe: it lets your opponent move twice and asks the engine how much that would have cost you. If a real threat existed (≥ 300cp) and you let it land, that is evidence against your prophylaxis; if it existed and you neutralised it, that is evidence for. The weight is derived from the threat's size, never from the language model. Probes are skipped when you were in check (the position would be illegal) and capped at six per game.

4.4 Bayesian Knowledge Tracing: how a score moves

Each skill holds a probability `pKnow` that you have mastered it; mastery shown on screen is simply round(`pKnow` $\times 100$). A new skill starts at `pKnow` $= 0.3$ (mastery 30). Each piece of evidence updates it with a weighted Bayesian step, using fixed parameters (slip 0.1, guess 0.2, learning transition 0.15).

■ **The update, in words**

A "for" observation raises `pKnow` toward the Bayesian posterior and adds a small learning bump; an "against" observation lowers it. **Weight scales the whole step**: a weight-1 blunder moves the number hard, a weight-0.3 inaccuracy nudges it, a weight-0 observation does nothing at all. Evidence is folded in sequence, so order within a skill matters.

The concrete effect of a *single* full-weight observation from a fresh skill (mastery 30):

Observation	New `pKnow`	New mastery
one ``for", weight 1.0	0.710	**71**
one ``against", weight 1.0	0.051	**5**
one ``against", weight 0.3	0.225	**23**
weight 0 (any)	0.300	30 (no change)

Confidence in the whole model grows with the number of samples, capped: $\min(95,\ \text{samples} \times 3)$ percent.

4.5 Own-mistake drills and SM-2 scheduling

Every mistake or blunder of yours that lost at least **100**cp — and for which the engine has a genuinely better move — becomes a drill. The correct answer is the engine's own best move; positions are de-duplicated so you never drill the same one twice.

Scheduling is SM-2-lite. A card carries an *ease* factor (start 2.0, floored 1.3, capped 2.6) and an interval that grows multiplicatively so a correct answer can never shorten it.

■ **The schedule**

Correct: ease rises (more for better-known skills), and the next interval is interval $\times$ ease, capped at **21 days**. **Incorrect:** ease drops by 0.2, the card returns in 5 minutes, and a lapse is recorded. After **4 lapses** a card is suspended as a "leech" and stops being scheduled. A card only counts toward your streak and retention when it was genuinely due — you cannot grind the same card for credit.

An answer is also accepted if it is any engine move within about 25cp of the best — two equally winning moves are both marked correct.

Correct answer #	Ease	Next interval	Meaning
1	2.08	2.08 days	graduates from ``new"
2	2.16	≈4.5 days	
3	2.24	≈10 days	
4	2.32	21 days (capped)	considered learned

Progression for a card on a mastery-30 skill, answered correctly each time.

5 A Complete Game of Play

We now run the whole system on one real game: Paul Morphy's "Opera Game" (Paris, 1858), analysed as though *you uploaded it and played Black* — the allied Duke and Count. Every number, verdict, and skill change in the boxes below is the **actual output** of the Master Chess pipeline run on this game's moves, not an illustration.

★ **THIS EXAMPLE WAS GENERATED, NOT HAND-WRITTEN**

The figures here come from running the app's real classifier, evidence rules, and Bayesian update on the stored analysis of this game. One consequence worth stating up front: the model's job is to read the moves honestly, and honest reading sometimes surprises — as it does here.

STEP 1 — UPLOAD & INGEST

You upload the game as Black. The ingester reads the header — `White: Paul Morphy`, `Black: Duke Karl / Count Isouard`, `Result: 1-0`, `ECO: C41`, `Opening: Philidor Defense` — matches your name to the Black seat, and replays all 33 plies, storing the position before and after each. The moves:

```
1.e4 e5 2.Nf3 d6 3.d4 Bg4 4.dxe5 Bxf3 5.Qxf3 dxe5 6.Bc4 Nf6 7.Qb3 Qe7 8.Nc3 c6 9.Bg5
b5 10.Nxb5 cxb5 11.Bxb5+ Nbd7 12.O-O-O Rd8 13.Rxd7 Rxd7 14.Rd1 Qe6 15.Bxd7+ Nxd7
16.Qb8+ Nxb8 17.Rd8#
```

A historical game carries no rating header, so — as §4 requires — no *level* or *plateau* is diagnosed from it. The skill evidence below is generated all the same.

STEP 2 — EVALUATE & CLASSIFY (ALL OF BLACK'S MOVES)

Stockfish scores every position; the classifier judges each Black move against the best available, phase-scaled. This is the complete, verbatim classification:

Move	Played	Phase	cp lost	Verdict
1…e5		opening	3	Best
2…d6		opening	9	Good
3…Bg4		opening	65	Inaccuracy
4…Bxf3		opening	54	Good
5…dxe5		opening	0	Best
6…Nf6		opening	67	Inaccuracy
7…Qe7		opening	18	Good
8…c6		opening	23	Good
9…b5		opening	84	Inaccuracy
10…cxb5		opening	139	Inaccuracy
11…Nbd7		middlegame	33	Good
12…Rd8		middlegame	58	Inaccuracy
13…Rxd7		middlegame	1	Best
14…Qe6		middlegame	132	Mistake
15…Nxd7		middlegame	mate	Blunder
16…Nxb8		middlegame	0	Best

Two things the real data teaches. First, the game is classified **opening** then **middlegame** throughout — the queen-aware phase rule keeps it out of the endgame despite the piece sacrifices. Second, the famous "losing" move **9…b5 is only an inaccuracy** (84cp). The move the engine actually flags as the **blunder is 15…Nxd7**, which walked into `16.Qb8+ Nxb8 17.Rd8#` — a forced mate (shown as "mate" rather than a centipawn figure).

STEP 3 — INFERRED EVIDENCE (EXACTLY WHAT THE RULES EMIT)

The rule engine reads those classifications. A crucial detail visible here: **inaccuracies emit no per-move skill evidence** — only clearly good moves (Best / Good) and clear errors (Mistake / Blunder) do. So Bg4, Nf6, b5, cxb5, and Rd8 produce nothing. What actually fires:

Skill	From (real moves)	Dir / wt
Opening principles	e5, d6, Bxf3, dxe5, Qe7, c6 (six sound opening moves)	for / 0.15
Tactical consistency	Nbd7, Rxd7, Nxb8 (sound middlegame)	for / 0.12
Tactical consistency	15…Nxd7 (middlegame blunder)	against / 1.0
Tactical pattern recog.	14…Qe6 (middlegame mistake)	against / 0.6
Tactical pattern recog.	15…Nxd7 (blunder pattern)	against / 0.5

Note what does *not* fire: no "conversion failure" (Black never reached +400), no "held-worse-position" credit (Black lost), and no resilience verdict (the run did not meet the sample conditions). The model records only what the game shows.

STEP 4 — THE PLAYER MODEL UPDATES (REAL BAYESIAN OUTPUT)

Each skill folds its evidence in sequence from a starting mastery of 30. The actual results:

Skill	Net evidence	Mastery	Trend
Opening principles	6 for	$30 \to 66$	↑
Tactical consistency	3 for, 1 against (wt 1)	$30 \to 12$	↓
Tactical pattern recognition	2 against	$30 \to 9$	↓

The surprise is honest and instructive: **opening principles rises to 66** even though Black *lost the game*. By cp-loss, Black's opening was sound — the defeat came from two middlegame errors, not the opening. A human eye remembers "the game where I got mated"; the model measures each phase on its own evidence. That is precisely the black-box the receipts exist to open.

STEP 5 — DRILLS HARVESTED

Two moves qualify as own-mistake drills — classified **Mistake** or **Blunder** and losing at least 100cp:

Position before	Skill	Correct answer
14…Qe6 (132cp)	tactical pattern recognition	the engine's best move
15…Nxd7 (mate)	tactical consistency	the engine's best move

Each becomes a four-choice card, due immediately, entering SM-2 scheduling. Note that `9…b5` and `10…cxb5` are *not* harvested — b5 lost under 100cp, and both were inaccuracies rather than outright errors.

STEP 6 — TRAINING PLAN PRESCRIBED

A skill needs at least 3 evidence samples to enter a plan. Here that admits *opening principles* (6 samples) and *tactical consistency* (4); with only 2 samples, *tactical pattern recognition* is honestly held back despite being the lowest score. Because the historical game has no rating, no plateau is diagnosed, so the plan uses its no-plateau hypothesis:

The hypothesis. *"Your weakest evidenced skills right now are tactical consistency, opening principles. No plateau pattern is diagnosable yet (needs a PGN-header rating), but these are the lowest-mastery skills we have real evidence for."*

Focus block 1 — Tactical consistency (mastery 12, 4 samples). *Matched reading* from the corpus: *Art of Combination* (Blokh), *Encyclopedia of Chess Combinations* (Chess Informant), *1000 Checkmate Combinations* (Henkin).

Focus block 2 — Opening principles (mastery 66, 6 samples), carrying its own matched titles. It ranks second only because it is the next evidenced skill — the plan is transparent about that rather than inventing urgency.

STEP 7 — THE ENGINE-GROUNDED NOTE

Finally the narrator is handed *only* verified facts — result, accuracy, blunder count, and the worst move (`15…Nxd7`, a mate-allowing blunder) — and asked for two to four sentences, then re-checked by the claim guard. A passing note reads like:

"Your opening was solid — it was the middlegame that cost you. 14…Qe6 let the pressure build, and 15…Nxd7 walked into a forced mate with 16.Qb8+ and 17.Rd8#. The habit to drill is the blunder-check before a recapture when your king is still in the centre."

Every move and claim there maps to the real record. Had the draft said "your 9…b5 was the losing blunder" — contradicting the classifier, which graded b5 an inaccuracy — the guard would reject it and you would see the deterministic summary instead. *(That is the exact error this manual's first draft made before it was checked against the engine.)*

★ **WHAT YOU WOULD DO NEXT**

Open *Drills* the next day: the `15…Nxd7` and `14…Qe6` positions are waiting. Choose the engine's move and the card schedules out; miss it and it returns tomorrow until the gap closes. That is the loop — diagnose, explain, prescribe — closing on your own moves.

6 Reference Card

6.1 Screens at a glance

Games	dashboard: stats + recent games
Upload	paste/drop PGN, live analysis
Game Review	per-move engine analysis, accuracy, summary
Player Model	level, plateau, 27 skills, evidence receipts
Training Plan	hypothesis + 3 focus blocks + books
Drills	SM-2 review of your own mistakes
Library	classic games + opening explorer
Account	GDPR export / delete

6.2 Numbers worth remembering

Skills / categories / levels / plateaus	27 / 4 / 7 / 5
Fresh skill mastery	30
Blunder threshold (mid / open / end)	300 / 350 / 200 cp
Drill harvest threshold	100 cp lost
Drill accept tolerance	∼25 cp of best
SM-2 ease (start / floor / cap)	2.0 / 1.3 / 2.6
Max review interval	21 days
Leech suspension	4 lapses
Plateau minimum samples	5
Plan minimum samples per skill	3
Free-tier analyses / day	5
Max games per upload	50

6.3 Classification legend

Best **Good** **Inaccuracy** **Mistake** **Blunder**